

# Understanding Dreams

*Lecture 2 — Do Dreams Have Meaning?  
Neuroscience, Psychology, and Your Own Dreams*

---

**Instructor:** Randolph S. Charlton, MD  
Adjunct Professor of Psychiatry, Stanford University School of Medicine

**Today:** How to work with your dreams redux, A look at where dreams originate

# Lecture 2 — Roadmap



Working with your dreams II — A dream from our class



A brief history of sleep science: Aserinsky, Kleitman, Dement



REM sleep and the question: what is the substrate of dreaming?



The “no meaning” view: Hobson, Crick, Zhang, Eagleman



The challenge: Mark Solms and what brain damage taught us



Three frameworks revisited: no meaning / intrinsic meaning / created meaning



More dreams: working with them together



Synthesis: what do we actually know?



What you can take from this into your own life

O P E N I N G

# More Dreams

---

*Dreams from our Last lecture & Last night · Discussion and exploration*

## A Patient's Dream — “Do You Love Me”

---

*I'm lying on the couch and you come down and lie down next to me. I say “you can't do that because you don't love me.”*

*You get up and sit in your chair and say “you're right, I don't love you.”*

*Then I realize that's a good thing — you're not supposed to love me. That's not the reason that I am here.*

*Freud's thoughts – the “you” is the therapist (as stand in for father)*

*What is the wish?*

*To be loved*

*What childhood calamity*

*Rejection, Loss of love*

*What is the emotional threat/pain?* *Abandonment anxiety, Shame (I am unloveable)*

*What are the defenses?*

*Reaction formation, Reversal, Intellectualization*

*Jung's thoughts – the “you” is the patient's internal self*

*What part of the dreamer is the “you” in the dream?*

*What does it mean if the dreamer doesn't love/isn't supposed to love himself?*

*What must the dreamer do to “balance this onesided situation?”*

## Dream from our class —

Someone or something is chasing me. I run into a classroom deep in the basement of a building. I open a door to find a boarded up baby changing room. I kick the boards open and find a huge gym-like room that is brightly painted in rainbow colors. The room has been abandoned for a while, and there is grime and black spots all over the walls. Particles float in the air. It smells strongly of paint.

### Freud's questions

Wish?

Childhood Calamity?

Emotional Pain/Threat?

Defences/Coping Strategy?

### Jung's questions

Who or what in the dreamer is driving him here?

Where in himself has the dreamer arrived?

Why have the rooms been abandoned?

What do the bright rainbow colors signify?

What are the internal black spots, grime & particles

What is the paint smell?

S E C O N D

# The Neuroscience of Dreaming

---

*A brief history of sleep science · What the brain is doing while you dream*

# A Very Brief History of Sleep Science — The Discovery of REM

1938

## First Observation

A. Jacobson notices: “When a person dreams... most often his eyes are active.” The eye movement during sleep is documented but not yet understood.

1953

## Dreams = REM

Aserinsky & Kleitman at the University of Chicago establish that dreams are the equivalent of REM sleep. For the first time, science can detect dreaming from outside the dreamer.

1957

## Awaking Dreamers

Bill Dement demonstrates that waking subjects during eye movements produces dream recall 90% of the time — vs. less than 10% during non-eye movement sleep. The experimental study of dreams becomes possible.

1958

## Activated Sleep

Dement describes “activated sleep” — eye movement, low-voltage EEG, absence of muscle activity. Eventually named REM sleep.

1959–65

## The Brainstem Connection

Jouvet destroys an area of the brainstem (the pons) in animals — REM stops. The machinery of REM sleep is located in the primitive, non-thinking brainstem.

1970

## The 90-Minute Clock

Hobson and colleagues show that brainstem cells fire rhythmically every ~90 minutes, powered by acetylcholine, generating REM as a preprogrammed biological process.

# What the Early Neuroscience Seemed to Conclude

**The apparent logic:** REM sleep is generated by random, rhythmic firing of primitive brainstem cells — the same cells that generate REM also generate dreaming. Therefore dreaming is a byproduct of brainstem activity, disconnected from the thinking, feeling brain. Therefore dreams are essentially random.

## Hobson's Activation-Synthesis (1977)

Dreams are narratives stitched together by the forebrain in an attempt to make sense of the meaningless activation of biochemical changes originating in the brainstem. The forebrain is doing its best — but the signal it is interpreting is **random noise**.

*Dreams are narratives, but not meaningful ones.*

## Crick's Reverse Learning Theory (1983)

Nobel Prize winner Francis Crick proposed that dreams are the brain's process of clearing out unnecessary or harmful memories — the **"garbage can theory."** The dream is what you see as the brain empties its trash.

*Dreams are meaningless fallout from memory housekeeping.*

## Zhang's Continual Activation Theory (2004)

Dreams are a byproduct of the process by which short-term **memories are transferred** to long-term storage. The imagery we experience is noise generated by that consolidation process.

*Dreams are the shadows cast by memory consolidation.*

## Eagleman's Defensive Activation Theory (2020)

REM sleep exists to keep the visual cortex from being colonized by neighboring cortical areas — keeping the visual system "active" so it doesn't get repurposed. Dream imagery is the **visual cortex firing to stay alive**.

*Dreams protect the visual brain — their content is incidental.*



T H I R D

# Mark Solms and the Challenge

---

*What brain damage taught us · REM ≠ Dreaming · A new possibility of where dreams come from*

# Mark Solms — A Different View (1990s–2020)

*Solms studied patients with specific brain damage. What he found was counter to the existing assumption of sleep science:*

## Finding 1: Brainstem damage doesn't stop dreaming

Patients with damage to the brainstem area that generates REM sleep — the area Jouvet had shown was essential for REM — still had dreams. If REM were the same as dreaming, this should be impossible.

## Finding 2: A different area stops dreaming

Damage to neurons nearby, but separate from the REM area — neurons belonging to a dopamine system connecting to the emotional and motivational brain — caused patients to lose the ability to dream entirely. REM sleep continued normally.

## The Conclusion: REM ≠ Dreaming

REM sleep and dreaming are simultaneous, but they are separate events. They are correlated — they tend to occur together — but they are not causally connected. The brainstem generates REM. Something else generates the dream.

## Where Dreams Actually Come From

The neural source of dreaming starts in the brainstem but connects to the emotional brain (the limbic system) and the motivational brain (the mesoparietal cortex). Dreams are generated by the feeling, motivated, wanting brain — not the reflexive brainstem.

# What Solms Changes — Why This Matters

*If dreams come from the emotional and motivational brain rather than from random brainstem noise, the implications are significant:*

**The neuroscientific case for no-meaning:** dreams are generated by random brainstem firing → they are processed by the forebrain → the “meaning” is just the forebrain making up a story about noise.

**Solms’ revision:** dreams are generated by the emotional and motivational brain → they reflect the brain’s needs, wishes, and unresolved emotional states. The “random noise” was never random.

## Freud revisited

Solms argues that his findings vindicate Freud’s core insight — that dreams are connected to wishes, needs, and emotional life — even if Freud’s specific theory of conflict and disguise needs revision.

## Consciousness and dreaming

The visual cortex combined with the emotional brain generates the vivid, felt quality of dream experience. Dreams are a form of consciousness — just a different kind than waking consciousness.

## Memory and emotion

The involvement of the emotional brain in dreaming is consistent with research showing that dreams preferentially process emotionally significant memories. The brain is not processing all memories equally.

## The debate continues

Solms’ work is contested. The relationship between REM, dreaming, emotion, and memory is genuinely complex and not yet fully resolved. The simple equation “REM = dreams = noise” is not necessarily accurate.

F O U R T H

# Three Frameworks Revisited

---

*No intrinsic meaning · Intrinsic meaning · Created meaning · The honest view*

# Three Frameworks for Dream Meaning — The Honest Accounting

*Each framework has genuine strengths. Each has genuine problems. Understanding all three is more useful than choosing one.*

## 1. Dreams Have No Intrinsic Meaning

### Strengths:

Grounded in neuroscience. Accounts for the bizarre, illogical quality of many dreams. Avoids the risk of false interpretation.

### Problems:

Assumes REM = dreaming (Solms has challenged this). Cannot explain why dream content is often emotionally charged and thematically coherent. Cannot explain why the same person repeatedly dreams the same themes.

## 2. Dreams Have Intrinsic Meaning

### Strengths:

Supported by Solms' neuroscience. Consistent with clinical experience across a century of psychotherapy. Dreams often do illuminate emotional life.

### Problems:

No way to prove that the meaning is intrinsic rather than created. The interpreter has enormous power — and historically this power has been abused. Which theory of meaning is correct: Freud, Jung, religious, or something else?

## 3. Meaning Is Created by the Dreamer

### Strengths:

Respects the dreamer's autonomy. Acknowledges that meaning is always partly constructed. Consistent with constructivist philosophy of mind.

### Problems:

The dreamer may create a comfortable, self-protective interpretation that avoids what the dream was actually pointing toward. Not all interpretive partners are equally trustworthy or skilled.

F I F T H

# More Dreams

---

*Working with them together ·*

## A Patient's Dream — “Go East” (39-year-old woman from NYC)

---

*I am in NYC in the back of a taxi. I am having an interesting conversation with the driver.*

*I want to go East at 9th Ave and Broadway. The road divides. 9th Ave goes straight. I want to turn left on Broadway and I tell him to go that way.*

*He goes straight on 9th and says, “It will be ok. Let’s go this way and see how it goes.”*

*I am VERY upset. I HAVE to go East.*

# The “Go East” Dream — Three Frameworks in Action

*The same dream, read through three different lenses. Notice what each framework reveals — and what it misses:*

## No-Meaning View

The emotional charge of the dream — the VERY upset, the sense of I HAVE to — is a byproduct of limbic system activity during REM. The story the patient tells about going East is the forebrain making narrative sense of arbitrary emotional activation.

---

*What this misses: why does this patient’s limbic system activate around control and direction? Why the same theme repeatedly?*

## Freudian View

The driver is the therapist. Going East may represent the direction the patient wants therapy to go. The driver’s calmly overriding her wish reflects her experience of the therapeutic relationship — and the conflict between her agenda and the therapist’s.

---

*The dream is working through an unconscious conflict about control, authority, and the fear of being led somewhere she doesn’t want to go.*

## Created Meaning

When this patient woke up panicked and said “I don’t understand — what’s so scary about a cab ride in Manhattan?” — she was beginning to create the meaning. The dialog with the therapist, exploring why going East mattered so intensely, is how meaning emerges.

---

*The meaning is real and useful — but it is created through the telling and the dialog, not retrieved from a fixed latent content.*



## Student Dream — Discussion 3

---

*[Insert student dream text here]*

**Apply all three frameworks:**

**No meaning:** What is the brainstem activation producing? **Intrinsic meaning:** What conflict or developmental message? **Created meaning:** What emerges in the telling?

## Dr. Charlton's Dream — “Practice in 10 Minutes”

---

*I walk up a gentle grassy hill on campus. I come upon a large group of young men sitting comfortably around talking and relaxing. I stop and sit down to join them.*

*I realize they are members of the Stanford football team and that I am included in the group. We discuss the season to come.*

*Someone calls out “practice will start in ten minutes.” Suddenly I’m aware that I am way too old to play football. I feel anxious about my age, limited strength and conditioning.*

*If I go to practice, I’ll be exposed and kicked off the team.*

# Working With Dr. Charlton's Dream — “Practice in 10 Minutes”

*This is an anxiety dream — one of the most universal types. Notice the structure:*

## **The wish:**

Belonging. Being part of the group. Being “included” on the Stanford football team — the prestige, the youth, the physical vitality.

## **The anxiety:**

The sudden awareness of age. “Way too old.” “Limited strength and conditioning.” The wish runs into the reality of the dreamer’s actual situation and introduces the possibility of rejection and banishment

## **The threat:**

Shame. Public exposure. Being “kicked off.” Not just failure, but witnessed failure — shame before others.

## **The defense:**

The dream doesn’t resolve the dilemma. The dreamer wakes before he has to go to practice — the classic anxiety dream ending.

## **The cross-cultural connection:**

This dream maps directly onto the anxiety dreams reported by both Chinese and Canadian students: “failing an examination,” “being too late.” The underlying structure — wish, threat, shame, exposure — is universal.

S I X T H

# What Do We Actually Know?

---

*An honest synthesis · What science and psychology agree on · What remains open*

# Take Home Synthesis — What We Can Say With Confidence

*Science and psychology do not agree on everything. But they agree on more than the debate suggests:*

## **Dreams are real neurological events**

Dreaming involves specific, identifiable neural systems — not random noise. These systems involve emotion, memory, and motivation.

## **Dream content is not random**

Dreams are thematically coherent across the night and across the life of the dreamer. The same emotional concerns recur. This is not what random noise looks like.

## **Dreams process emotional experience**

The evidence that dreams preferentially involve emotionally significant material — and that dream deprivation impairs emotional regulation — is robust across multiple research traditions.

## **Meaning is created in the waking encounter with the dream**

Whatever a dream contains intrinsically, the meaning that is therapeutically and personally useful is always created in the process of remembering, telling, and reflecting. The dreamer is an active participant.

## **Dreams can illuminate what waking life conceals**

The clinical evidence across a century is consistent: bringing dreams into conversation often opens material — fears, wishes, conflicts — that the patient could not approach directly.

## **You cannot know for certain what a dream means**

Interpretive humility is required. A plausible interpretation is not necessarily a correct one. The interpreter has power — and that power can be abused. The dreamer's own associations are the most trustworthy guide.

# What Remains Genuinely Open

**Does the brain generate specific meaning in the dream itself, or does meaning emerge in the waking/narrating process?**

This is an unresolved question. Solms thinks the brain generates meaning. Hobson thinks the forebrain creates it post-hoc. We don't yet have the tools to answer it definitively.

**Does it Matter?**

If the "meaning" of a dream is created by the dreamer, sometimes with therapeutic help, does it matter where the meaning comes from? Perhaps not to the dreamer.

**Are the Freudian specific mechanisms — condensation, displacement, wish-fulfillment, denial, dissociation, reaction formation, undoing, ect. — neurobiologically generated or intellectual add ons?**

The clinical observations are robust. The neural mechanisms are still being mapped. The concepts have proven clinically and personally useful regardless of their ultimate neurological substrate.

**Why do some people remember dreams vividly and others rarely?**

The process of laying down short term memory during sleep and dreaming is under study. Clinically, awakening memories are evanescent and fade quickly.

Dream recall varies enormously between individuals and across the lifespan. The causes are not fully understood. Motivation to remember appears to play a role.

**Is there a relationship between dreaming and creativity?**

Yes. The evidence that the dreaming brain makes novel connections is unequivocal. Kekulé discovered the benzene ring from a dream. Paul McCartney heard the melody of Yesterday in a dream.

The mechanisms of dream formation resemble those of the creative process in waking life.

# What You Can Take From This Into Your Own Life

*You don't need to be a therapist to learn something real from your dreams. You need only to pay attention.*

- 1 Write them down:** Dreams fade within minutes. A notebook or phone beside the bed changes everything. You don't have to understand the dream — just record it.
- 2 Name the emotion first:** Before you try to interpret anything, notice what you felt in the dream and what you feel remembering it. The emotion is your most reliable guide.
- 3 Associate freely:** Don't analyze. Say whatever comes to mind about each significant image. The meaning tends to emerge through the associations, not through reasoning about the dream.
- 4 Look for the recurring theme:** A single dream is interesting. A series of dreams with the same emotional concern is clinically significant. What does your dreaming mind keep returning to?
- 5 Hold interpretations lightly:** A plausible interpretation is a hypothesis, not a fact. The dream may mean something you haven't yet thought of. Stay curious longer than feels comfortable.
- 6 Notice what the dream doesn't show you:** Dreams sometimes omit things, soften things, or change endings. The veterinarian changed the ending for his wife. The gap or distortion often holds as much information as the content itself.

“  
*Dreaming permits each and every one of us to be quietly and safely insane every night of our lives.*

---

— *William Dement, MD*

*The question we have been exploring is whether that nightly insanity is also, in some sense, wisdom — a language the deeper mind uses when ordinary speech is not enough.*

---

**The answer is: probably yes — but only if you pay attention to it.**